Department of Engineering Science
School of Science & Technology



CES 552 Network Architecture
Internet Protocol

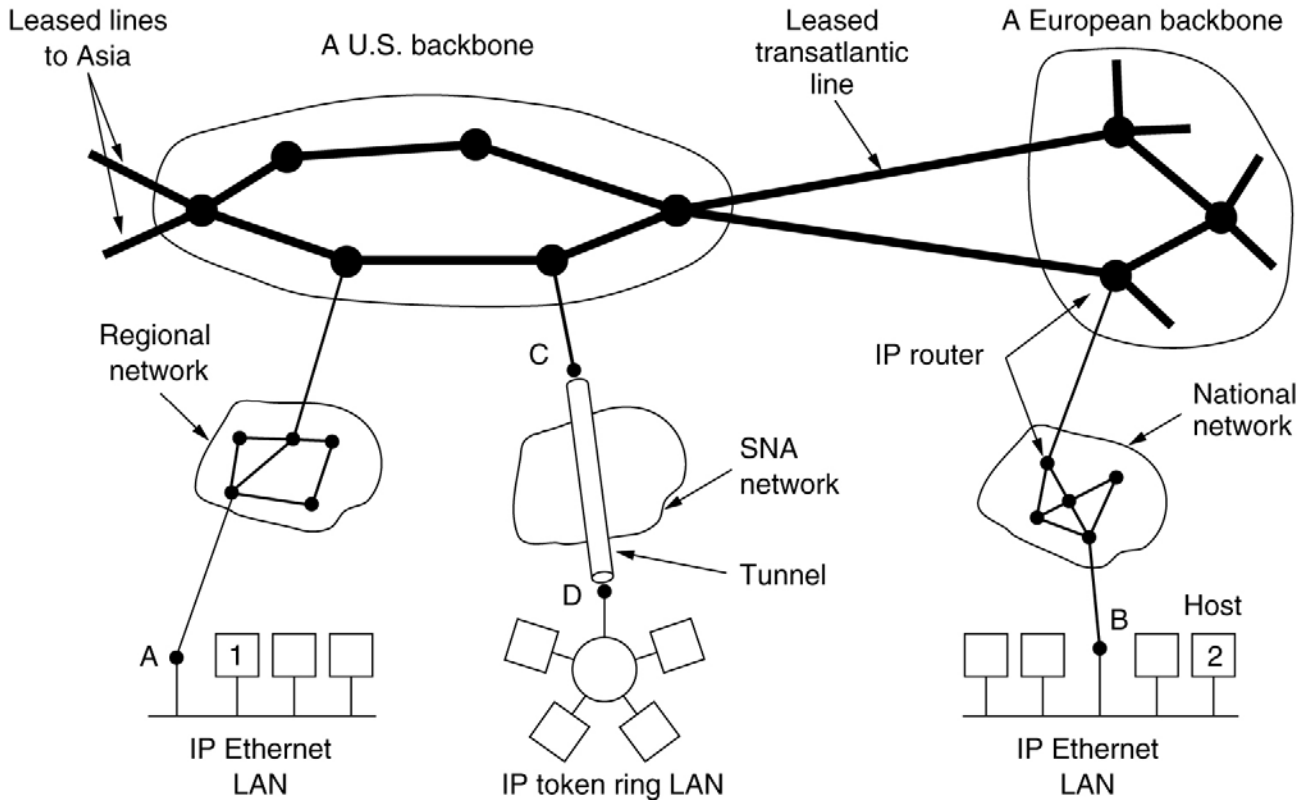
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OUTLINE

- **Internet Protocol**
- **IP address**
- **Internet Control Protocol**
- **Routing in Internet**
- **IPv6**

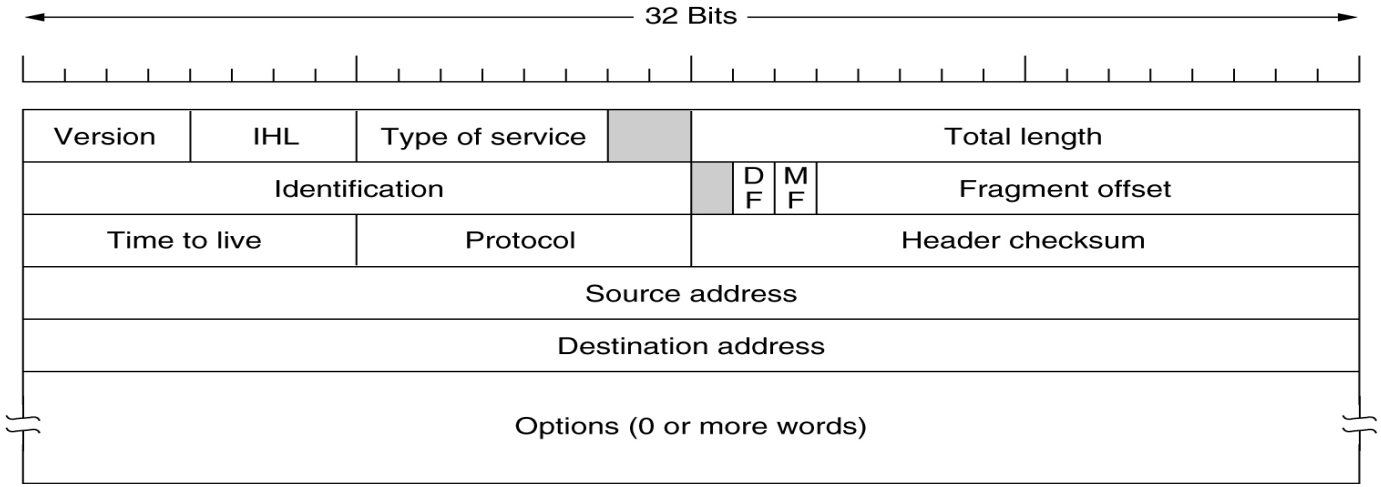
IP

- **Internet Protocol (IP) is the glue for Internet**
  - IP was designed with internetworking in mind.
  - It provides best-efforts (not guaranteed) way to transmit datagram from source to destination.



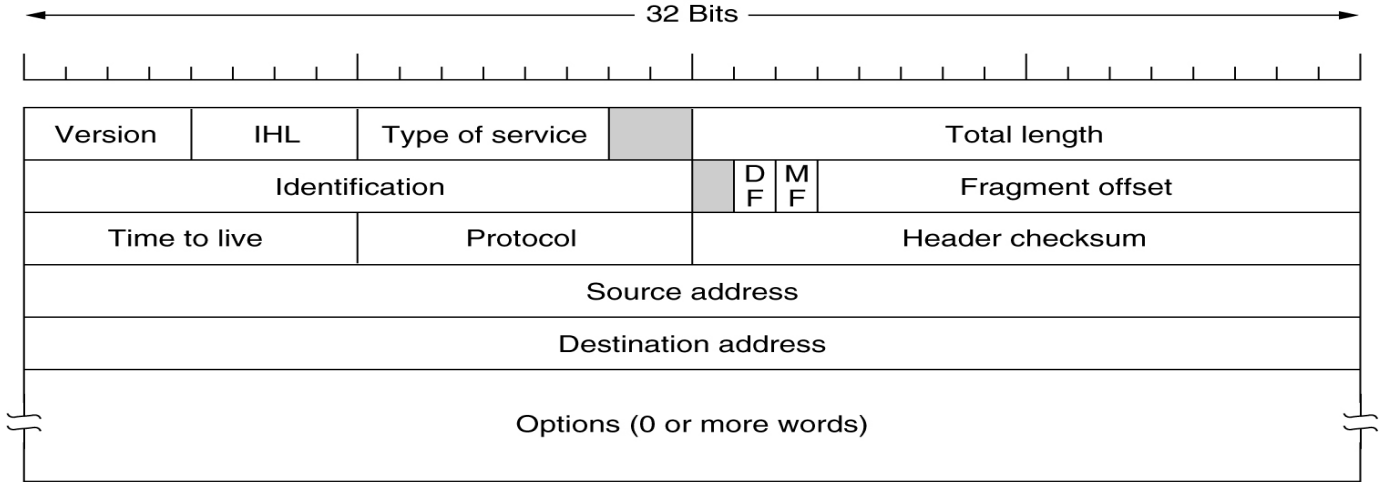
SNA: System Network Architecture

IP: PDU



- Version: the version of the protocol (IPv4, IPv6)
- IHL (4 bits): header length (in the unit of 32-bit word, or 4-byte word).
  - Min value = 5 ➔ 5 x 4 = 20 bytes. Max value = 15 ➔ 15 x 4 = 60 bytes.
- Type of service (6 bits): used to distinguish different service classes (QoS).
  - Differentiated services: 4 queuing priorities, 3 discard probabilities, etc.
  - This field has been ignored by routers.
- Total length (16 bits): the total length of the IP packet (including header and data)
  - Maximum length: 2^{16} = 65536 bytes = 64KB

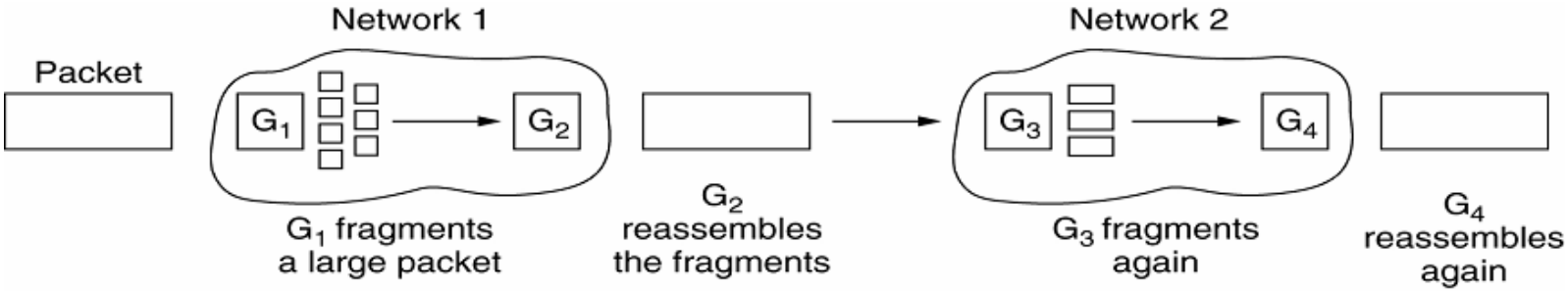
IP: PDU



- Identification (16 bits): identify which datagram a fragment belongs to.
  - Fragments belonging to the same datagram has the same ID field.
- DF: Don't fragment.
- MF: More fragment. More fragments are following.
  - All fragments except the last one have this bit set to 1.
- Fragment Offset (13 bits): the seq. # of the first elementary fragment in the current fragment. Elementary fragment: 8 bytes.
  - 2^{13} = 8192 elementary fragments. ➔ 8192 x 8 = 65536 bytes = 64KB.

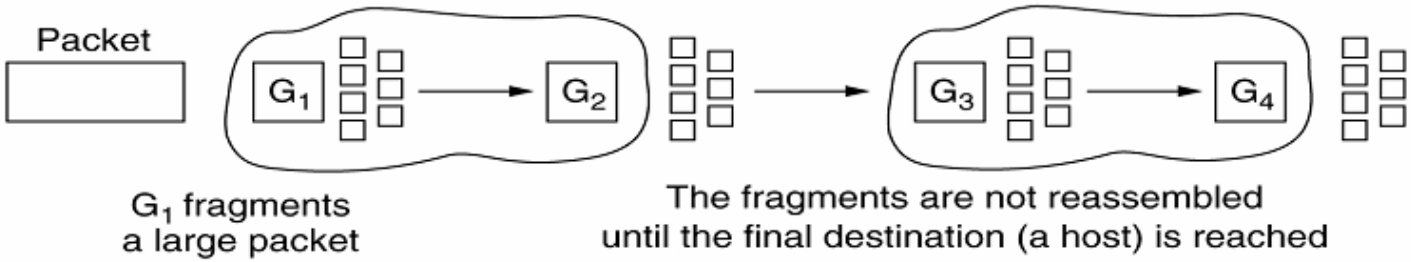
IP: FRAGMENTATION

- **Why fragmentation**
  - different networks have different packet size limitations.
- **Transparent fragmentation**
  - Fragmentation/reassembling are performed at intermediate gateways.



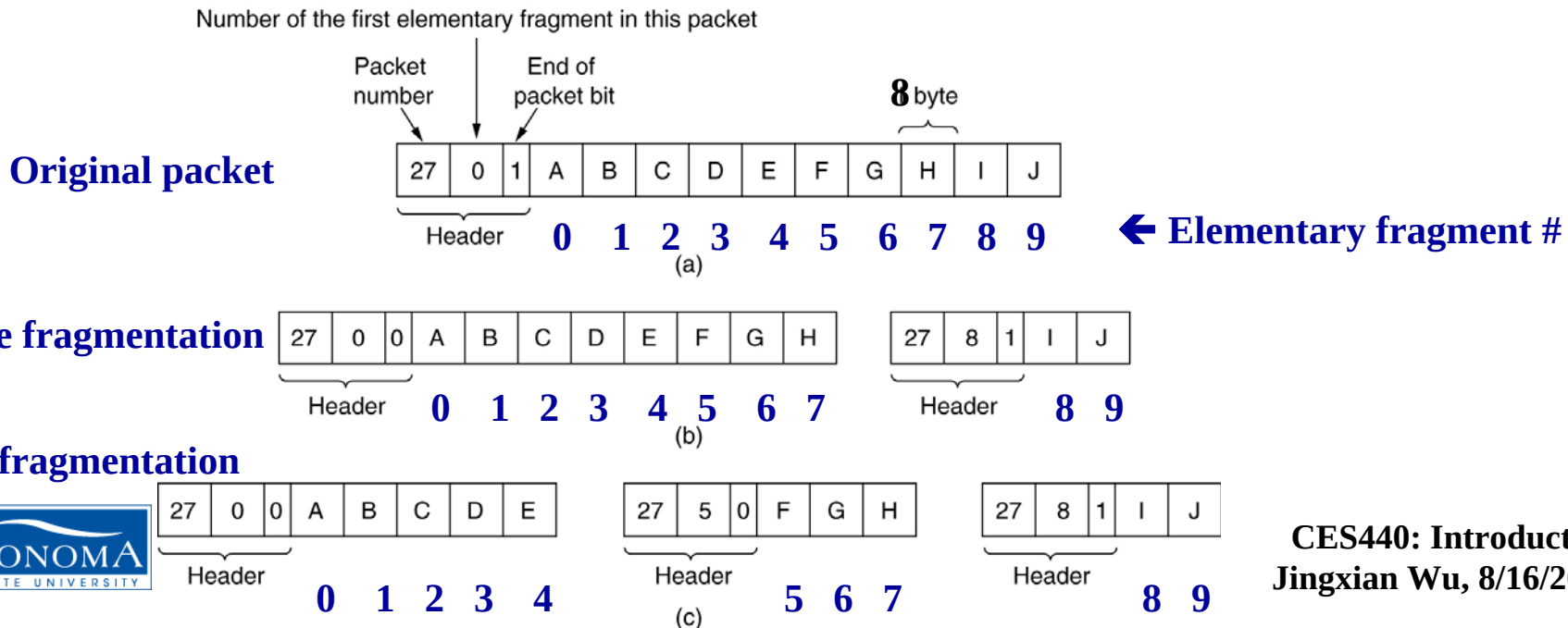
- **Non-transparent fragmentation (Used in Internet)**
  - Reassembling are performed at destination host.

(a)

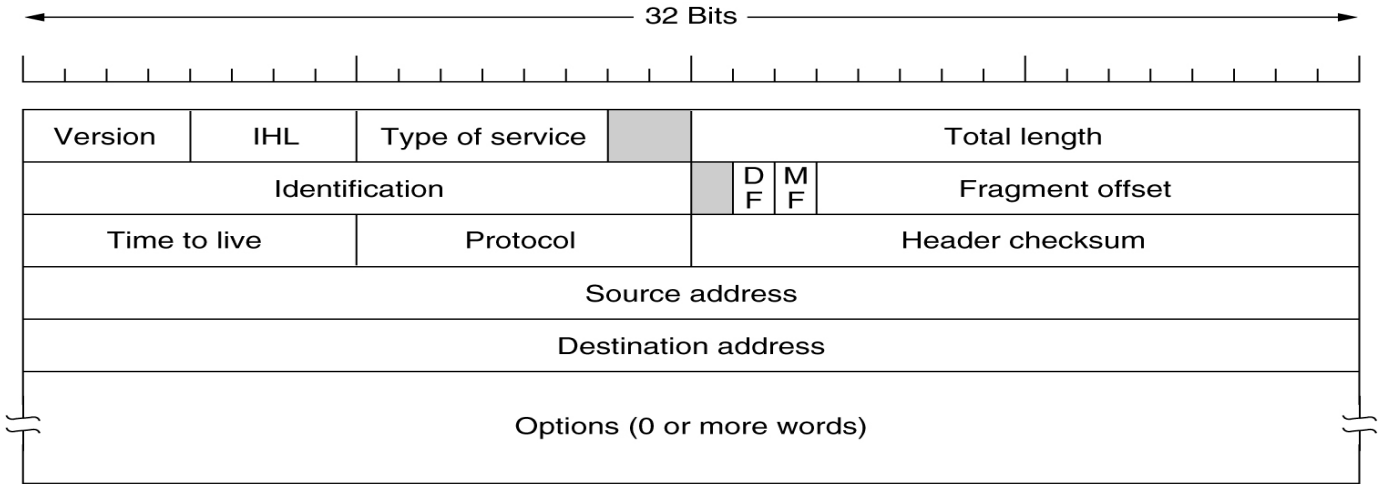


IP: FRAGMENTATION

- **How to number the fragment for non-transparent fragmentation?**
  - Define an elementary fragment size small enough for all networks (e.g. 8 bytes)
  - Each fragmentized packet contains several elementary fragments.
  - Each fragmentized packet must have the following three fields
    - Packet # (the original packet #)
    - Elementary fragment # (the seq. # of the first elementary fragment in this packet).
    - End of packet bit (to indicate if this is the last fragmentized packet of the original packet.



IP: PDU



- Time to live (8 bits): decrease by 1 after each hop.
  - When hits 0, packet is discarded and a warning is sent back to source.
- Protocol: which protocol is on the transport layer (TCP, UDP, etc.)
- Header checksum: check header only. MUST be recomputed at each router! (Why?)
- Source address (32 bits): IP address of source.
- Destination address (32 bits): IP address of destination.

IP: OPTIONS

- **Option: variable length. Can contain many options**
  - Each option starts with 1 byte word to identify the option
  - Some options

Option	Description
Security	Specifies how secret the datagram is
Strict source routing	Gives the complete path to be followed
Loose source routing	Gives a list of routers not to be missed
Record route	Makes each router append its IP address
Timestamp	Makes each router append its address and timestamp

- More options can be found at www.iana.org/assignments/ip-parameters

IP: OPTIONS

- **Example: record route**

00000111	length	pointer	route data

Record the route of this packet

– 00000111: the word identifying this is a record route option.
– Length: the total length of this option (including everything)
– Pointer: pointing into the route data indicating the first byte of the next area (4-byte) to store an address
  • Smallest value: 4 (corresponding to the first byte in route data.
– Route data: a series of IP addresses that are on the route of the packet.

OUTLINE

- Internet Protocol
- **IP address**
- Internet Control Protocol
- Routing in Internet
- IPv6

IP ADDRESS

- **IP address: 32-bit**
  - 2^{32} = 4295 million addresses.
  - Format: dotted decimal.
    - Binary: 11000000.00101001.00000110.00010100
    - Hexadecimal: C0.29.06.14
      - Hexadecimal:Binary

        0: 0000; 1: 0001; 2: 0010; 3: 0011

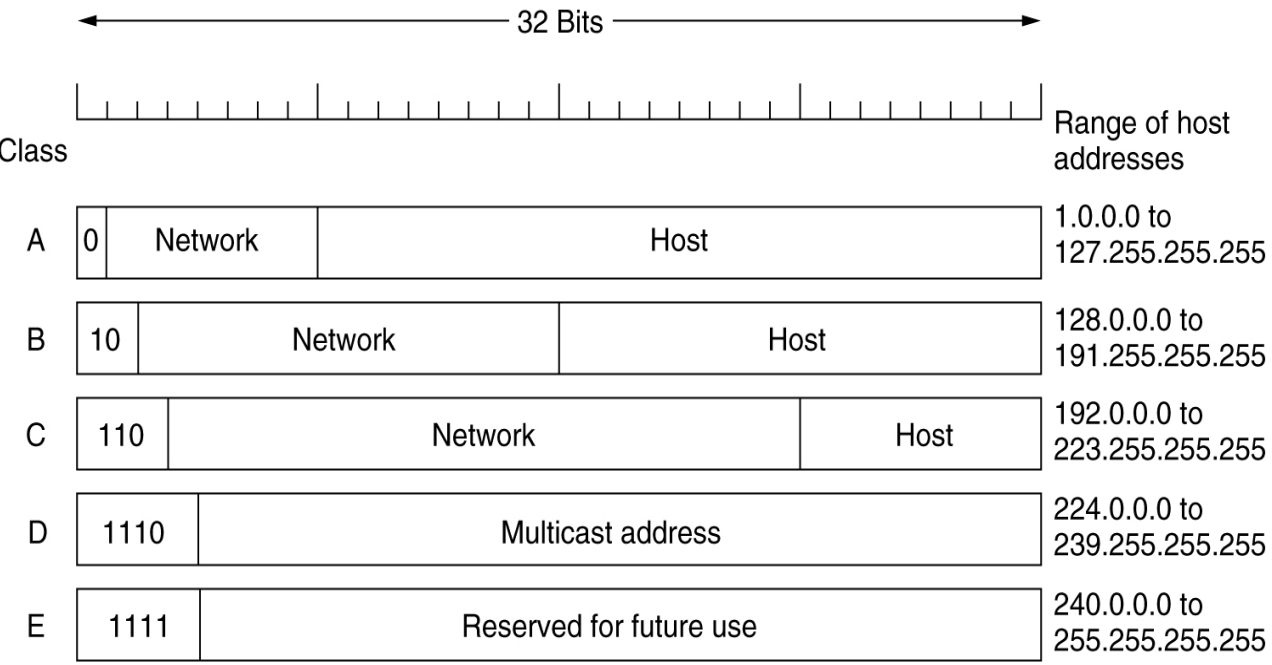
        4: 0100; 5: 0101; 6: 0110; 7: 0111

        8: 1000; 9: 1001; A: 1010; B: 1011

        C: 1100; D: 1101; E: 1110; F: 1111
    - Decimal: 192.41.6.20
      - 12x16+0 = 192; 2x16+9=41; 0x16+6=6; 1x16+4=20.
  - IP address assignment is managed by ICANN (Internet Corporation for Assigned Names and Numbers)
  - Usually contains two fields: network address + host address

IP ADDRESS

- **IP addresses are divided into 5 classes**
  - 0.0.0.0: this host
  - 255.255.255.255: broadcast on the local network

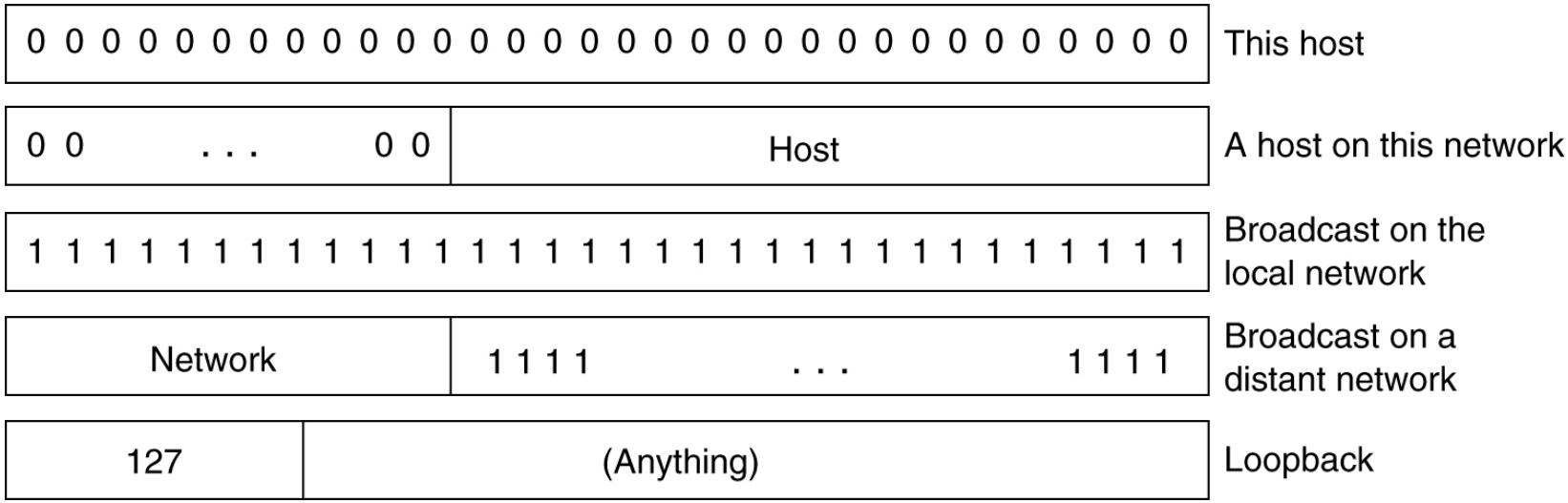


- 128 networks with 16 million hosts
- 16,384 networks with 64K hosts.
- 2 million networks with 256 hosts.
- Multicast
- Reserved

E.g. 130.157.5.73➔10000010.10011101.00000101.01001001

IP ADDRESS

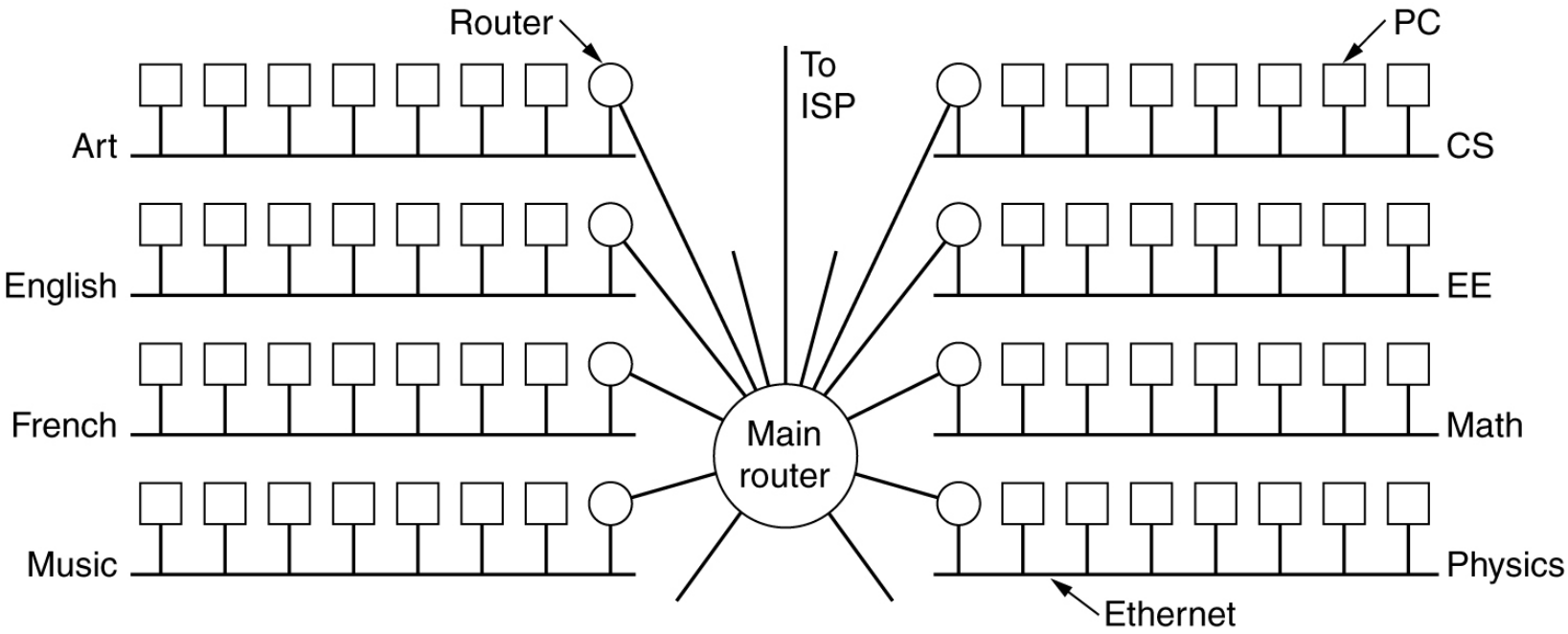
- **Special IP addresses**



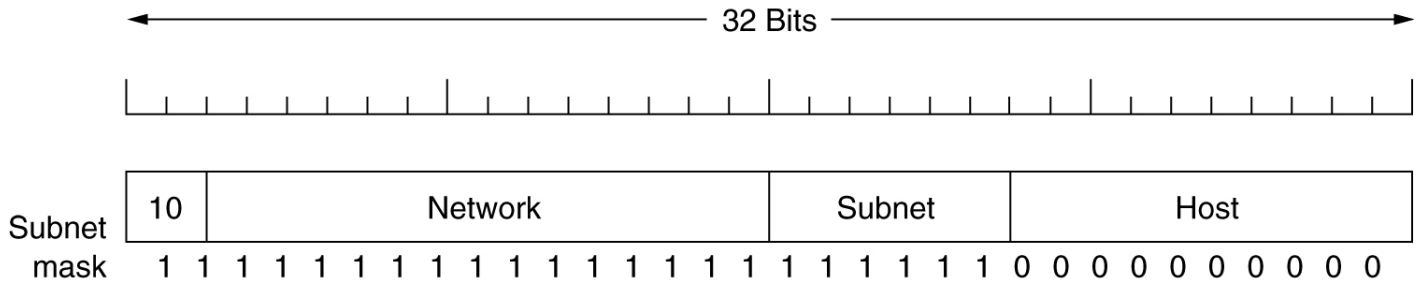
  - Loopback: for testing only.
    - Packets sent to this address will not be put in the wire.
    - They are going to be processed locally as incoming packets.

IP ADDRESS: SUBNET

- **Subnet (Different from the collection of routers and Tx Lines!)**
  - Allow a single network be split into small parts for internal use but still act like a single network to the outside.
    - E.g. Each LAN within the network is a subnet.



IP ADDRESS: SUBNET



- **Subnet (Cont'd)**
  - Some bits of the host number are used to identify subnet.
    - E.g. in a class B network, 16-bits are used to identify hosts in the network. Among the 16-bit, 6 bits are used for subnet # (64 Ethernets), 10 bits are used for host # (1022 hosts in each subnet).
    - Subnet mask: 11111111.11111111.11111100.00000000 (255.255.252.0, or 130.157.23.16/22)
      - 22 bits are used for network # and subnet #; 10 bits are used for host #
  - Router in this network can route packet based on the subnet #.

IP ADDRESS: SUBNET

- **Subnet Example**

subnet 1: 100000010 00110010 000001|00 00000001
subnet 2: 100000010 00110010 000010|00 00000001
subnet 3: 100000010 00110010 000011|00 00000001

subnet mask:
11111111 11111111 111111|00 00000000
Hexadecimal representation:

Dotted decimal representation:

of subnet:
of hosts/subnet

IP ADDRESS: CIDR

- **CIDR (Classless Inter Domain Routing)**
  - Motivation: many IP addresses are wasted by the class scheme. (Class B)
  - Assign IP address to an organization by giving the actual # of IP requested by it and a network mask.

University	First address	Last address	How many	Written as
Cambridge	194.24.0.0	194.24.7.255	2048	194.24.0.0/21
Edinburgh	194.24.8.0	194.24.11.255	1024	194.24.8.0/22
(Available)	194.24.12.0	194.24.15.255	1024	194.24.12/22
Oxford	194.24.16.0	194.24.31.255	4096	194.24.16.0/20

  - Each organization is assigned a base address, a last address, and a mask.

	Address	Mask
C:	11000010 00011000 00000000 00000000	11111111 11111111 11111000 00000000
E:	11000010 00011000 00001000 00000000	11111111 11111111 11111100 00000000
O:	11000010 00011000 00010000 00000000	11111111 11111111 11110000 00000000

Each IP contains two fields: network address (specified by mask), host address.
All computers belonging to same network has same network address!

IP ADDRESS: CIDR

- **CIDR (Classless InterDomain Routing)**
  - Two level hierarchical routing
  - Each router contains all the network addresses and corresponding mask.
  - When an packet comes in, its destination address is ANDed with one mask.
    - Check the network address (specified by the mask)!
    - If the AND result is the corresponding base address for that mask, then the packet is forwarded to that network.
    - If more matches are found, choose the one with the longest mask.
  - E.g. 194.24.17.4 (11000010 00011000 00010001 00000100)
  - AND Cambridge: 11000010 00011000 00010000 00000000
  - AND Edinburgh: 11000010 00011000 00010000 00000000
  - AND Oxford: 11000010 00011000 00010000 00000000 (Winner!)

	address	Mask
C:	11000010 00011000 00000000 00000000	11111111 11111111 11111000 00000000
E:	11000010 00011000 00001000 00000000	11111111 11111111 11111100 00000000
O:	11000010 00011000 00010000 00000000	11111111 11111111 11110000 00000000

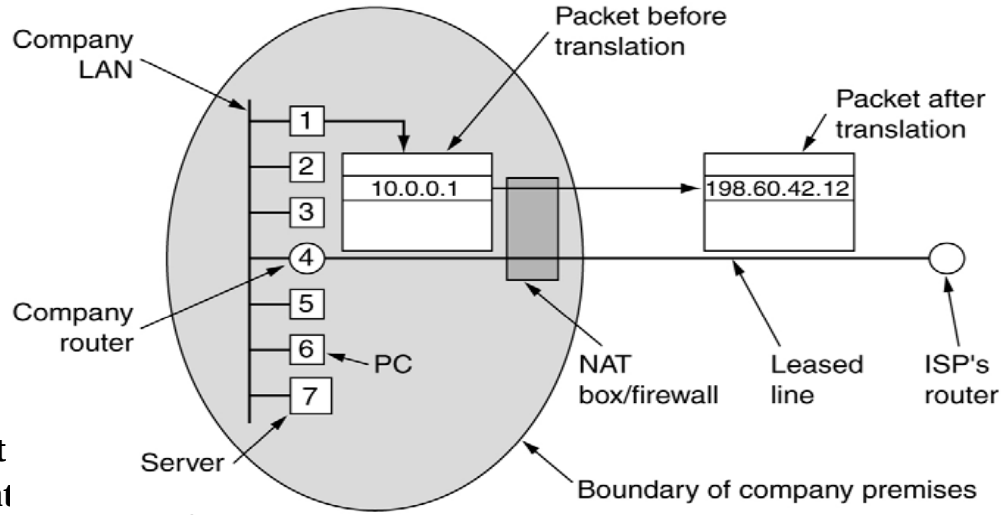
IP ADDRESS: NAT

- **NAT (Network Address Translation)**
  - Motivation: Most users want static IP address ➔ We are running out of IP addresses!
  - Solution: Assign 1 or a few IP to company or ISP; Private IP are used inside company or ISP; NAT is used for address translation!
    - Private IP address ranges

      10.0.0.0 - 10.255.255.255/8 (16,777.216 hosts)

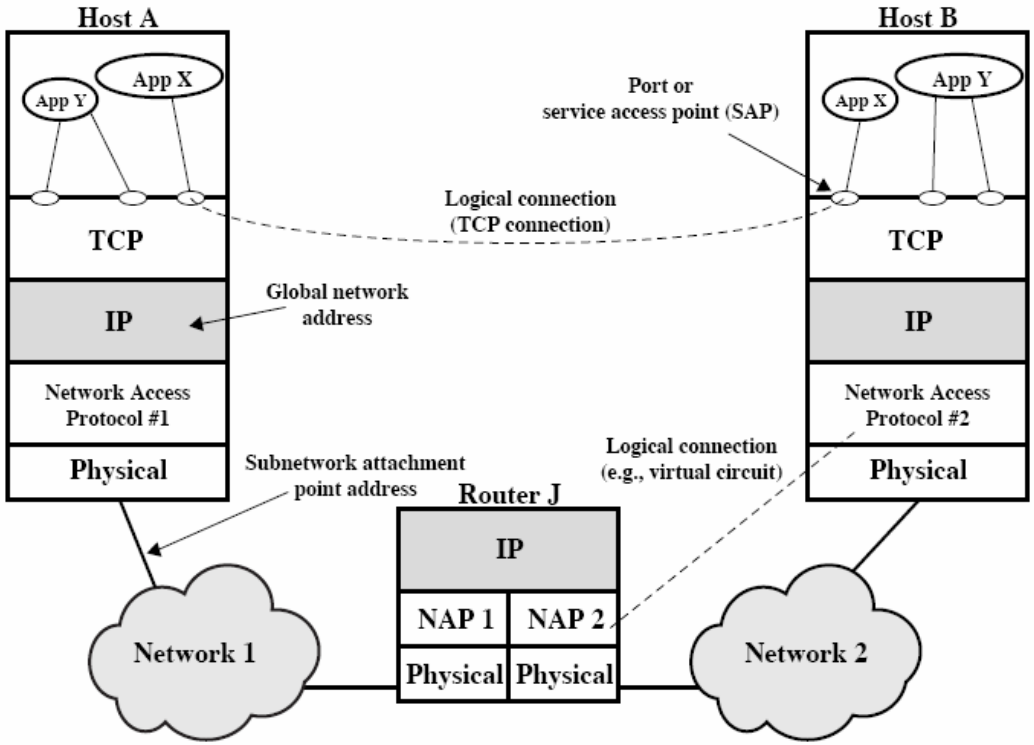
      172.16.0.0 - 172.31.255.255/12 (1,048,576 hosts)

      192.168.0.0 - 192.168.255.255/16 (65,536 hosts)
    - Packets leaving local network will be replaced with true IP address.



IP ADDRESS: NAT (CONT'D)

- All incoming packets are addressed to the real IP of the network.
  - How to distinguish incoming packets for different hosts?
- TCP (or UDP) has header fields for source port and dest. port.
  - Each port corresponding to one of the processes in host.
    - E.g. port 80: http; port 21: ftp;



IP ADDRESS: NAT (CONT'D)

- NAT uses TCP source port to distinguish hosts!
  - Each host is assigned a unique TCP port # at NAT.
  - NAT maintains a table (host # and original port # v.s. NAT mapping port #)
  - When a packet is sent to NAT from the private network
    - its IP address is replaced by real IP address;
    - its TCP src port is changed to the NAT port based on (host # and original TCP port #).

IP ADDRESS: NAT (CONT'D)

Example: computer A (10.0.0.1) ← private addresses

NAT (198.60.42.12) ← public address
A mapping (10.0.0.1, 2000) ←→ (4000) is recorded in NAT

A has an FTP session with remote server 130.157.0.1

A → server	Source IP	Source port #	Dest. IP	Dest. Port #
Original packet	10.0.0.1	2000	130.157.0.1	21
After NAT				

A ← server	Source IP	Source port #	Dest. IP	Dest. Port #
Original packet				
After NAT				

OUTLINE

- Internet Protocol
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- **Internet Control Protocols**
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INTERNET CONTROL PROTOCOL

- **Internet control protocol (ICP)**
  - Control the operations of network layer
  - Examples
    - ICMP (Internet Control Message Protocol)
    - ARP (Address Resolution Protocol)
    - RARP (Reverse Address Resolution Protocol)
    - DHCP (Dynamic Host Configuration Protocol)
- **The ICP information is carried in the payload field of IP packets**
  - They are still considered network layer protocols, because they are closely related to network layer operations
    - E.g. assign network address by DHCP

ICP: ICMP

- **Internet Control Message Protocol (ICMP) - RFC 792**
  - Carried in the payload field of an IP packet

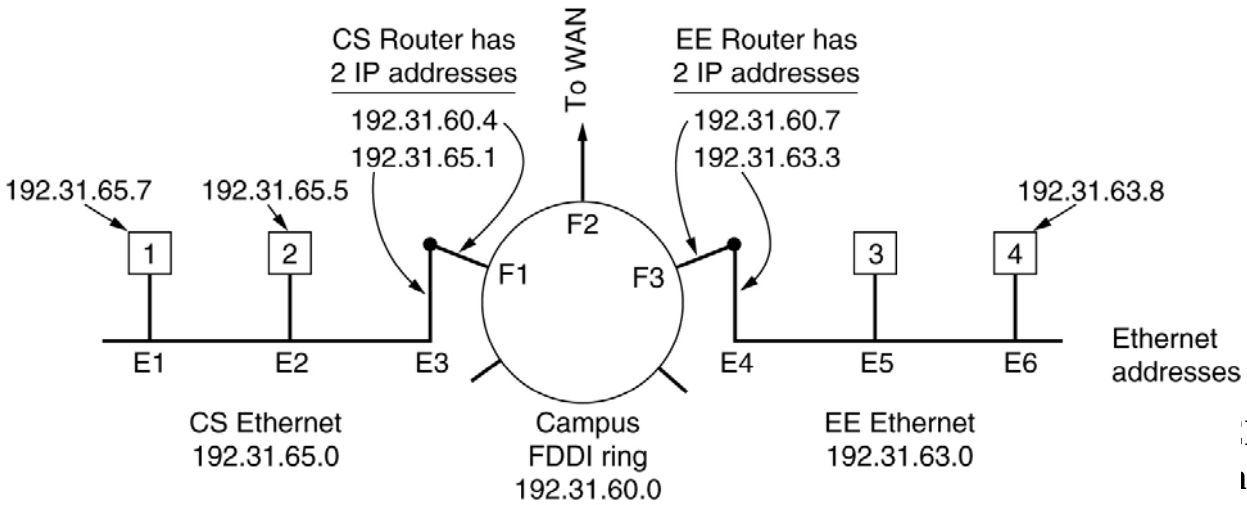
ping

Message type	Description
Destination unreachable	Packet could not be delivered
Time exceeded	Time to live field hit 0
Parameter problem	Invalid header field
Source quench	Choke packet
Redirect	Teach a router about geography
Echo request	Ask a machine if it is alive
Echo reply	Yes, I am alive
Timestamp request	Same as Echo request, but with timestamp
Timestamp reply	Same as Echo reply, but with timestamp

Source quench: choke packet for congestion control.

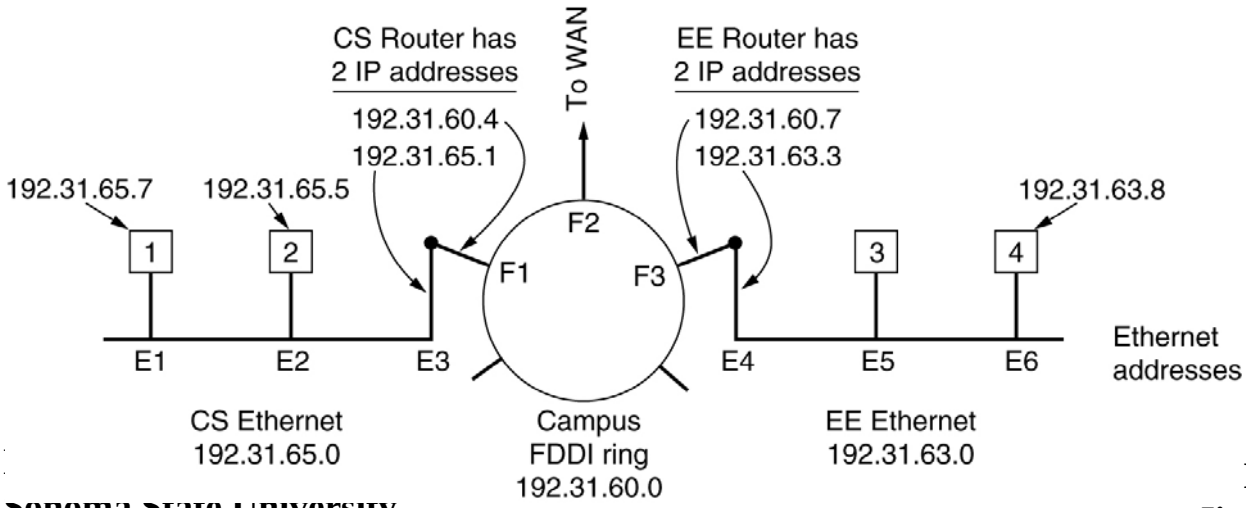
INTERNET: INTERNET CONTROL PROTOCOLS

- **Address Resolution Protocol (ARP): how to find Ethernet address (MAC) by IP address? (RFC 826)**
  - MAC address: hardware address in the data link layer.
    - Every machine in the world has its unique MAC address
  - IP address: address for the network layer. Can be reconfigured.
- **Example 1: E1 ➔ E2**
  - Host 1 broadcasts an ARP packet asking: what is the MAC address for 192.31.65.5?
  - Host 2 answers with its own MAC address: the MAC for 192.31.65.5 is E2
  - All the hosts make an entry (192.31.65.5, E2) and (192.31.65.7, E1) in their own (IP, MAC) mapping table for host 1 and 2.



INTERNET: INTERNET CONTROL PROTOCOL

- **ARP Example 2: E1 to E6**
  - Host 1 uses ARP finds out (192.31.63.8, E3) . Host 1 sends a data packet (192.31.63.8, E3)
    - E3 is the point handling all remote traffic
  - F1 rips the IP packet out of the frame. It encapsulate the packet in a new MAC frame (192.31.63.8, F3)
  - F3 uses ARP find out the MAC address of E6, and encapsulate the IP packet in a new MAC frame (192.31.63.8, E6)

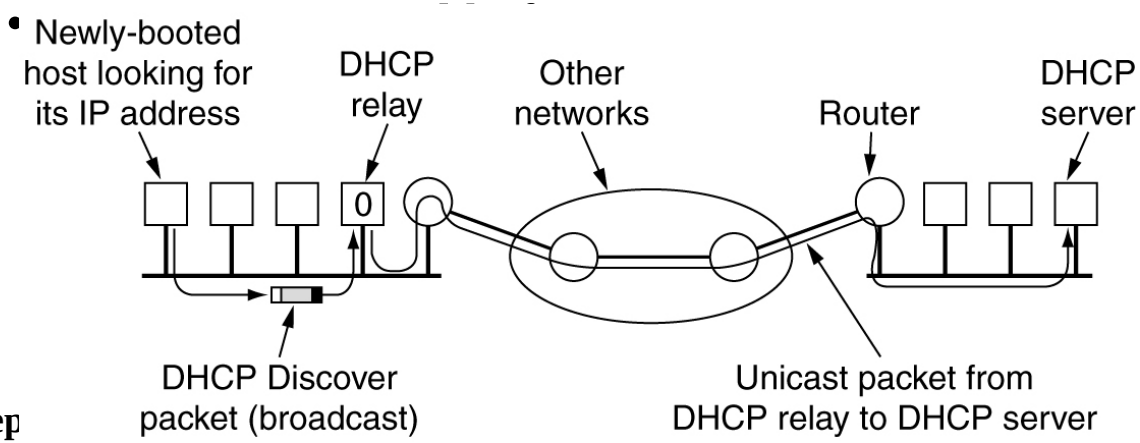


INTERNET: INTERNET CONTROL PROTOCOLS

- **RARP (Reverse Address Resolution Protocol) (RFC 903)**
  - A newly booted host broadcast its MAC address, and asking for IP address.
  - RARP server replies with the IP by looking the MAC in its table (preconfigured by network administrator).
  - Router won't forward MAC broadcast message (dest: all '1's).
    - Each network needs an RARP server.
- **BOOTP (RFC 951)**
  - A newly booted host broadcast its MAC address using UDP messages
    - UDP message can be forwarded by routers
  - BOOTP server will reply with IP and other information
    - E.g. the IP address of file server to download Operating System image.
  - The MAC/IP mapping table must be configured by hand.

INTERNET: INTERNET CONTROL PROTOCOLS

- **DHCP (Dynamic Host Configuration Protocol) (RFC 2131)**
  - A newly-booted host broadcasts a DHCP DISCOVER packet.
  - Each LAN has a DHCP relay agent
    - Forward the message to DHCP server through unicast.
    - The DHCP server might not be reachable through broadcasting.
  - The DHCP DISCOVER packet reaches the DHCP server
  - DHCP server automatically assigns an IP address to the host.
  - Leasing: IP is assigned for that host for a period of time.

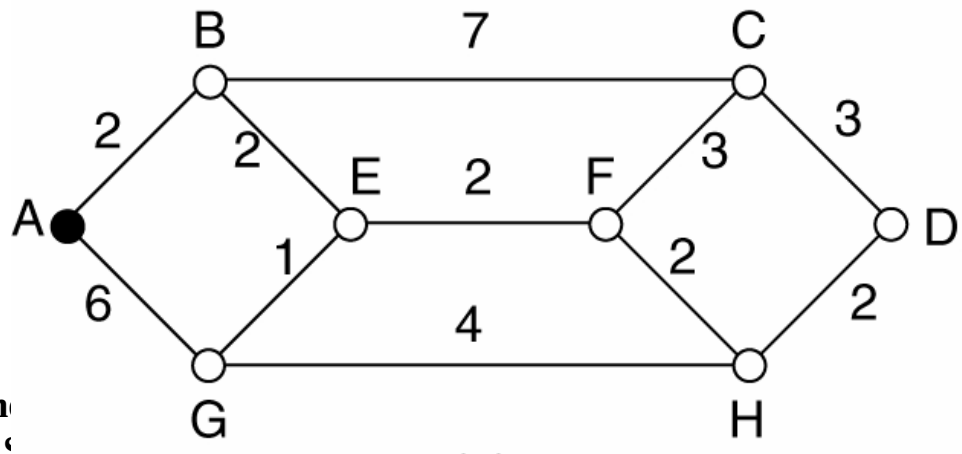


OUTLINE

- Internet Protocol
- IP address
- Internet Control Protocols
- **Routing in Internet**
- IPv6

ROUTING: GENERAL CONCEPT

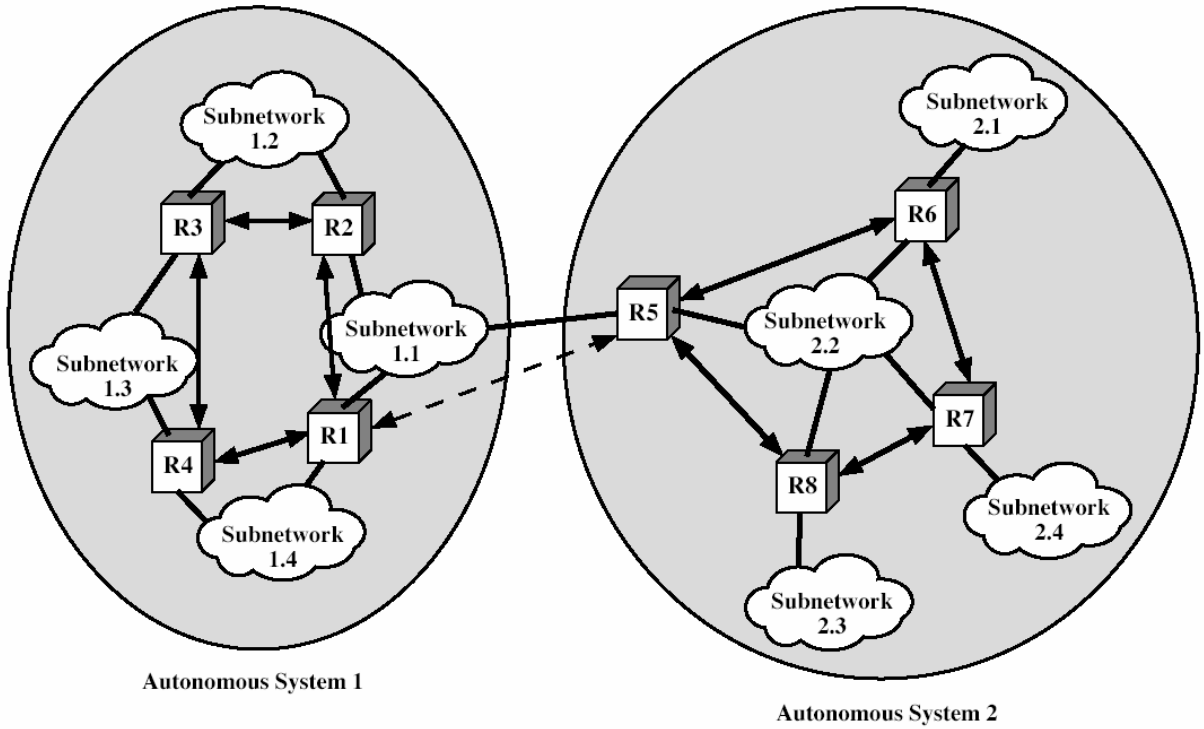
- **Routing Algorithm**
  - Deciding which output line an incoming packet should be forwarded to.
  - Datagram: routing is performed for every packet. (Internet)
  - VC: routing is only performed at connection setup ➔ session routing.
- **Find the route with the smallest cost**
  - Different measures can be used to measure "cost"
    - E.g. # of hops, physical distance (km), queuing delay …
  - In practical system, the "distance" measure is usually a function of the distance, bandwidth, average traffic, communication cost, mean queue length, delay, and other factors.



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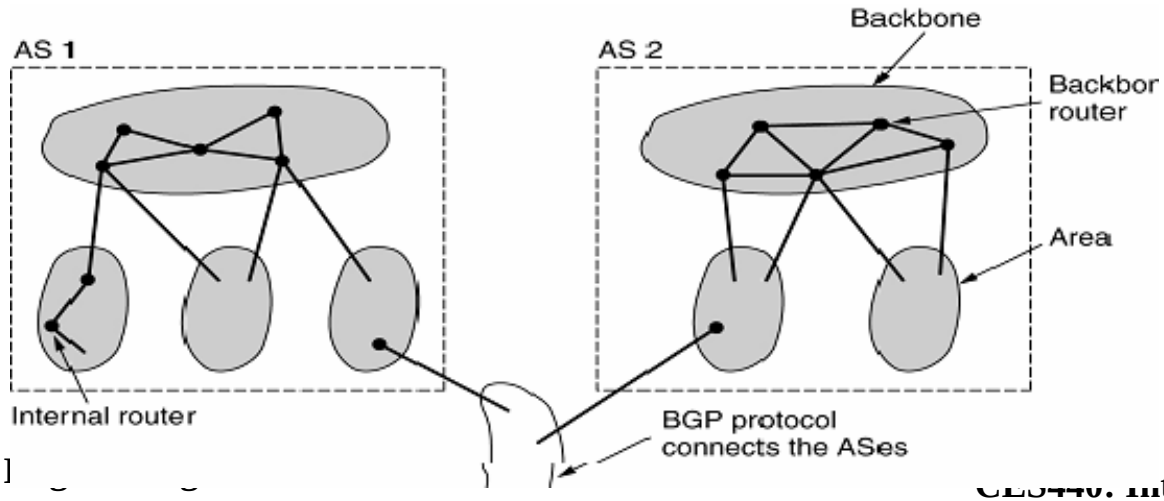
ROUTING: INTERNET

- **Concept: Autonomous system**
  - A set of routers and networks with a common routing protocol
  - Except failure, these is at least one path between any pair of nodes belonging to one AS ➔ it's fully connected.
- **Routing inside AS: interior gateway protocol (IGP)**
- **Routing among ASes: border gateway protocol (BGP)**



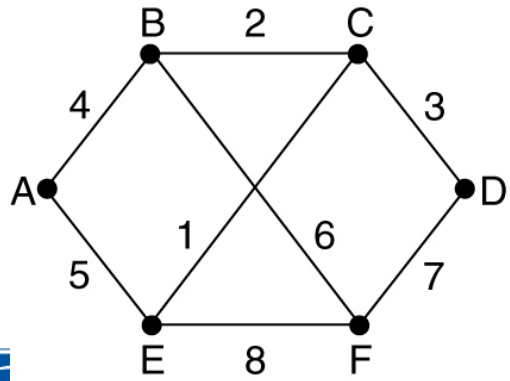
ROUTING: OSPF

- **OSPF (Open Shortest Path First) – RFC 2328**
  - An interior gateway (multi-protocol router) routing protocol:
  - AS is divided into areas: a network or a set of contiguous networks.
    - Each AS has a backbone area connected to all other areas.
    - Router connecting to two or more areas is part of backbone.
  - Four types of routers:
    - internal routers, area border router, backbone router, AS boundary router



ROUTING: OSPF

- **OSPF uses Link State Routing (5 steps). For each router:**
  - 1. Learning about neighbors
  - 2. Measuring distances to all neighboring routers
  - 3. Build link state packet (LSP)
    - {Source ID, Seq#, Age, (neighbor ID, distance to neighbor)}.
  - 4. Periodically distribute the link state packet through flooding
    - LSPs are only flooded within its area (not the entire AS)
    - After collecting LSPs, each router will have full knowledge of its area.
  - 5. Find the shortest path.



(a)

Link State Packets

A		B		C		D		E		F	
Seq.		Seq.		Seq.		Seq.		Seq.		Seq.	
Age		Age		Age		Age		Age		Age	
B	4	A	4	B	2	C	3	A	5	B	6
E	5	C	2	D	3	F	7	C	1	D	7
		F	6	E	1			F	8	E	8

(b)

ROUTING: OSPF

- **OSPF (cont'd)**
  - Each router has full knowledge of the topology in its own area.
    - Also knows the path to different areas.
  - Backbone routers have full knowledge of backbone topology.
    - The best path for each destination area.
- **Intra-area v.s. Inter-area**
  - Intra-area: find the shortest to destination based on topological knowledge.
  - Inter-area: (backbone connecting all areas)
    - the packet is forwarded to backbone from area border router of the source area
    - Backbone forward the packet to destination area.
    - The border router of the destination area forward the packet to destination host.

ROUTING: BGP

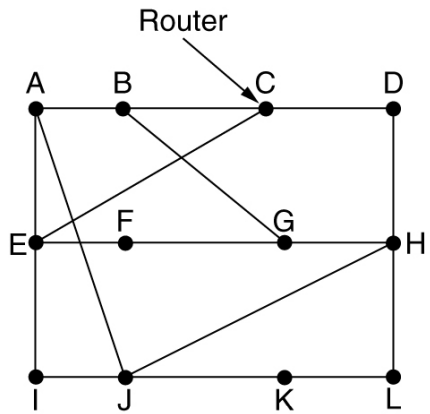
- **BGP (Border Gateway Protocol)**
  - An exterior gateway routing protocol: routing between ASes.
  - The major difference from interior gateway routing: need to consider politics
    - E.g. Do not use United States for traffic from British Columbia to Ontario;
    - Traffic starting at IBM should not pass Microsoft;
    - Transit traffic of only paid customers, etc.
    - Never put Iraq on a route for a packet starting from Pentagon
  - The rules are manually configured in BGP routers.

ROUTING: BGP

- **BGP used an enhanced distance vector routing protocol**
  - Each router maintains a routing table containing one entry for each destination AS as:
    - (destination AS, distance to destination, complete route to destination).
    - In original distance vector routing: (destination AS, distance to destination).
    - The complete route to destination is used to avoid count to infinity problem (Tanenbaum p. 357-360)
  - Each router exchanges information only with its immediate neighbors.
    - In link state route, the link state packet is distributed to all routers in the area.
  - Router finds out the best route to destination based on information from neighbors.

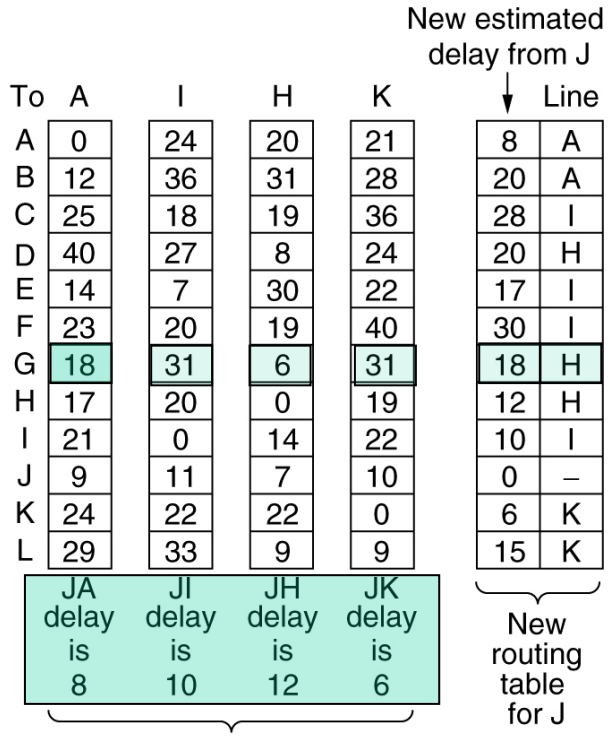
INTERNET: DISTANCE VECTOR ROUTING

- **Distance Vector Routing**
  - Each router maintain a routing table (a vector) contains one entry for each destination router as: (destination, distance to destination, which line to use).
  - The contents of the tables are dynamically updated by exchanging information with neighboring routers.
  - Example: How router J computes its routing table.



(a)

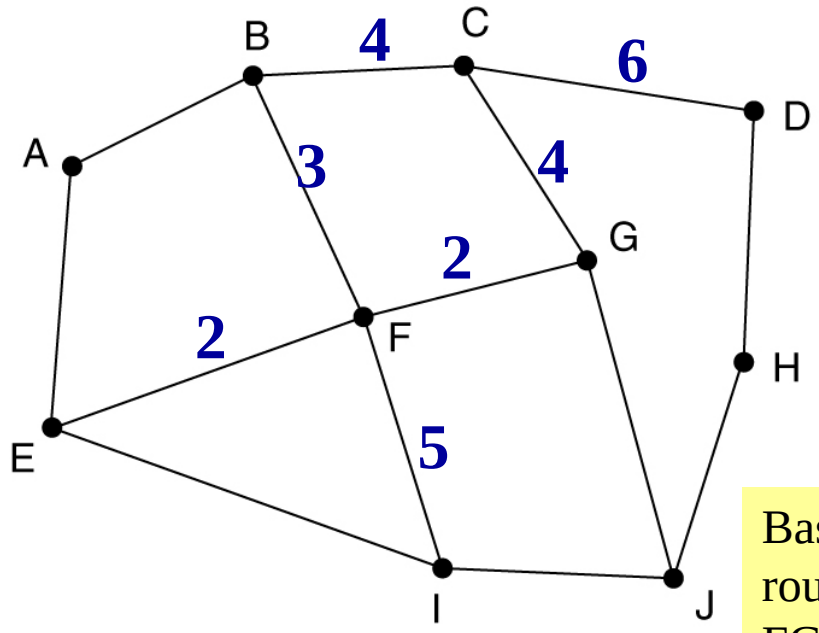
J→A: 8, A→G: 18. J→A→G: 8+18=26.
J→I: 10, I→G: 32. J→I→G: 10+32=42.
J→H: 12, H→G: 6. J→H→G: 12+6=18.
J→K: 6, K→G: 31. J→K→G: 6+31=37.



(b)

INTERNET: BGP

- **BGP Example**



(a)

Information received by F about destination D

Information F receives from its neighbors about D

From B: "I use BCD" cost is 10
From G: "I use GCD" cost is 10
From I: "I use IFGCD" cost is 17
From E: "I use EFGCD" cost is 14

Based on information from all neighboring routers, F will record the route to D as FGCD with cost 10+2 = 12.

(b)

If FG is broken, F will know information from I "IFGCD" and information from E "EFGEC" is no longer valid since FG is on the route.

OUTLINE

- Internet Protocol
- IP address
- Internet Control Protocols
- Routing in Internet
- **IPv6**

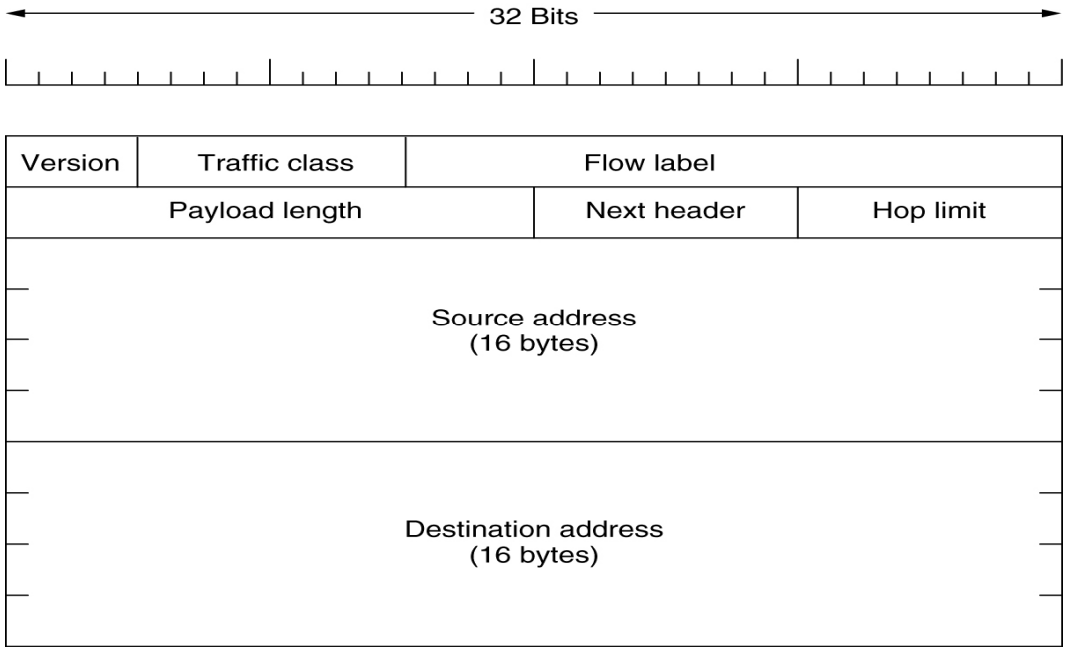
IPV6

- **IPv6: the evolution of IP**
  - 16 bytes of addresses: support effectively unlimited addresses.

$$2^{16\times 8} = 3.4\times 10^{38}$$

    - For the entire earth (water & land), allow 7×10^{23} per square meter!
  - Header is simplifier compared with IPv4
    - Fixed length: 40 bytes
    - Packets can be processed faster in router➔lower delay, larger throughput.
  - Better support of options
    - Many required fields in IPv4 become options in IPv6
  - Better security
  - Better support of QoS.
  - Not compatible with IPv4, but compatible with most other protocols
    - TCP, UDP, ICMP, IGMP, OSPF, BGP, DNS

IPV6: MAIN HEADER



- Version: 6
- Traffic class: different service classes for QoS
- Flow label: the label of a particular connection. Routers can use this field for QoS purpose. (e.g., this flow is delay sensitive).
  - Each flow is designated by: (source, destination, and flow #).

IPV6: MAIN HEADER

- **Main Header (Cont'd)**
  - Payload length: how many bytes following the 40-byte header
  - Next header: which of the 6 optional header is following the main header.
    - If no optional header, tells which transport protocol to pass the packet to.
  - Hop limit: same as Time to live in IPv4.
  - IPv6 address notation: eight groups of four hexadecimal digits

    8000:0000:0000:0000:0123:4567:89AB:CDEF
    - Can be simplified as

      8000::123:4567:89AB:CDEF
      - Consecutive zeros are replaced by ::
      - Leading 0s are omitted.

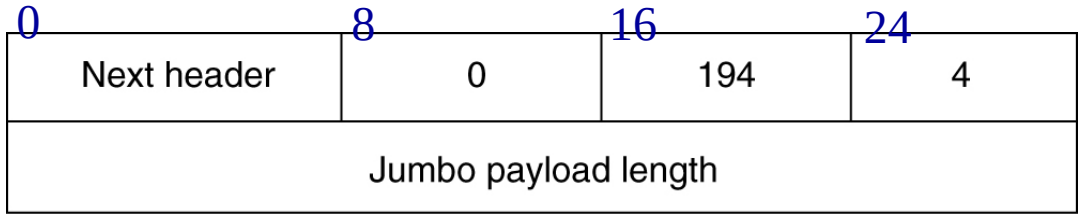
IPV6: EXTENSION HEADERS

- **Currently 6 optional headers defined.**

Extension header	Description
Hop-by-hop options	Miscellaneous information for routers
Destination options	Additional information for the destination
Routing	Loose list of routers to visit
Fragmentation	Management of datagram fragments
Authentication	Verification of the sender's identity
Encrypted security payload	Information about the encrypted contents

IPV6: EXTENSION HEADERS

- **Example: hop-by-hop extension for jumbograms**
  - Jumbogram: the datagram of size of several gigabytes
  - Used for supercomputer applications



  - Next header: the type of the optional header immediately following this one
  - 0: the length of this optional header excluding the first 8 bytes.
    - For this header, the total length is 8 bytes.
  - 194: this is an extension header for jumbograms
  - 4: the "Jumbo payload length" is specified in the unit of 4-byte word.

For example: if jumbo payload length is 2^20 = 1048576, the payload length of the jumbogram is: 1048576 x 4 = 4194304 Bytes

4194304/1024 = 4096 MBytes ~ 4.096 GBytes